

**CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY**

Sixth Semester of B. Tech. (EC) Examination

December 2013

EC 310 Optical Communication

Date: 05.12.2013, Thursday

Time: 1:30 p.m. to 4:30 p.m.

Maximum Marks: 70

**Instructions:**

1. The question paper comprises of two sections.
2. Section I and II must be attempted in separate answer sheets.
3. Make suitable assumptions and draw neat figures wherever required.
4. Use of scientific calculator is allowed.

**SECTION – I**

- Q - 1 (a) Describe with a neat and labeled sketch the construction of surface emitting LED [04]  
(b) Compare single mode fiber and multimode fiber. [03]
- Q - 2 (a) Enumerate the different attenuation mechanisms in fiber. Explain any one in detail. [05]  
(b) Explain the key elements of optical fiber communication system with diagram. [05]  
(c) Explain three key transition processes involved in LASER action. [04]

**OR**

- Q - 2 (a) Explain Fabry Perot resonator cavity for LASER diode [05]  
(b) Explain the terms power launching, coupling efficiency. [05]  
(c) A step index fiber has a core refractive index of 1.480, a core radius equal to  $4.5\mu\text{m}$  and core cladding difference is 0.25 percent. What is the cutoff wavelength for the fiber? [04]
- Q - 3 (a) Derive the expression for the power launched from the surface emitting LED source to step index and graded index fiber for radius of the source greater than and less than the core radius. [07]  
(b) Describe operating principal of OTDR and explain OTDR trace with figure. [07]

**OR**

- Q - 3 (a) What are the possible lensing schemes used to improve optical source to fiber coupling efficiency? Discuss with necessary diagram. [05]  
(b) With suitable diagram explain the cutback technique for attenuation measurement. [05]  
(c) Briefly explain mechanical misalignment in fiber. [04]

## SECTION – II

- Q - 4 (a) Define Following term: [03]  
(i) Critical angle (ii) Numerical aperture (iii) Dispersion in fiber
- (b) Explain quantum efficiency and responsivity of a photodiode. [04]
- Q - 5 (a) Explain PIN photo detector with structure, principle of operation and the performance. Compare it with RAPD. [05]
- (b) Explain digital signal transmission in optical communication with figure. [05]
- (c) A multimode silica fiber has a core refractive index  $n_1=1.48$  and a cladding refractive index  $n_2=1.46$ . Calculate (i) numerical aperture and (ii) maximum acceptance angle for the fiber. [04]

### OR

- Q - 5 (a) Explain principle of operation, the structure and the characteristics of APD. [05]
- (b) Write a short note on optical receiver. [05]
- (c) A double-hetrojunction InGaAsP LED emitting at a peak wavelength of 3110 nm has radiative and nonradiative times of 30 ns and 100 ns, respectively. The drive current is 40 mA, Find (a) the bulk recombination and (b) the internal quantum efficiency. [04]
- Q - 6 (a) Describe Mach-Zehnder interferometer with required sketch and equations. [06]
- (b) Explain 2x2 fiber coupler and also discuss different losses associated with it. [06]
- (c) What do you mean by population inversion? [02]

### OR

- Q - 6 (a) Draw and explain basic structure of SONET Frame. [06]
- (b) Write a short note on Fiber Grating Filters. [06]
- (c) What are the key system features of WDM? [02]

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